Data analyst or data scientist: what can you reach in 6 months?

Short answer: six months is enough to become job-ready as a data analyst. It is enough to learn the foundations of data science, but not usually enough to be hired as a data scientist.

Data analyst

Realistic in 6 months at about 10 to 15 hours a week. Good first target.

Data scientist

Foundations in 6 months. Job-ready usually takes 12 to 24 months, often through an analyst role first.

The two roles side by side

Titles vary by company. Treat this as the typical split.

	Data analyst	Data scientist
Main question	What happened, and why?	What will happen, and what should we do about it?
Daily work	Pull data with SQL, clean it, build reports and dashboards, explain findings to teams.	Build and test models, run experiments, write production-quality code, work with large or messy data.
What you deliver	Dashboards, reports, recommendations.	Models, predictions, experiments, data products.

Math needed	Descriptive statistics, percentages, basic probability.	Probability, inferential statistics, linear algebra, some calculus.
Core tools	Excel, SQL, Tableau or Power BI, a little Python.	Python, SQL, scikit-learn, Jupyter, Git, often cloud tools.
Typical entry	Any degree plus a portfolio. Many career changers start here.	Often a degree in a quantitative field, a master's degree, or experience as an analyst.
Hiring bar	Lower and more entry-level roles available.	Higher and fewer true entry-level openings.

Skills you need

Start with the shared basics, then choose a path.

Basic skills (both roles)

- Curiosity and asking clear questions about data
- Spreadsheet comfort: formulas, pivot tables, lookups
- Statistics basics: mean, median, spread, correlation, distributions
- Communication: explaining numbers to people who are not technical
- Data cleaning habits: spotting duplicates, gaps, and errors
- Business thinking: tying an analysis to a decision

Data analyst **Data scientist**

Technical skills

- **SQL**: joins, grouping, window functions

Proof for employers

- 3 portfolio projects with a written summary of findings

- **Excel**: advanced formulas, pivot tables, charts
- **BI tools**: Tableau or Power BI dashboards
- **Python basics**: pandas, matplotlib
- **Statistics**: hypothesis tests, confidence intervals

- One dashboard that others can open and use
- A clean GitHub or portfolio page
- One project from a real, messy dataset

Can you do it in 6 months?

The plan assumes about 10 to 15 hours a week. Tap a month to open the skills behind it.

Analyst plan Scientist plan

Month 1	Python and statistics. Learn pandas and core probability.
Month 2	SQL and data cleaning. Work with messy, real datasets.
Month 3	Math for ML. Cover linear algebra and calculus basics.
Month 4	Machine learning basics. Regression, trees, and model evaluation.
Month 5	Two end-to-end projects with clear write-ups.
Month 6	Review gaps, then apply to analyst or junior data roles while you keep learning.

How close does 6 months get you?

Data analyst (job-ready, about 6 months)

Data scientist (job-ready, about 12 to 24 months)

Your time depends on your starting point. If you already know Excel, some programming, or statistics, you move faster. If you are starting from zero, plan for the long end of each range.

Best route for most people: aim for a data analyst role in 6 months, then grow toward data science on the job.

Timelines are typical estimates, not guarantees. Hiring varies by country, company, and market.